Solution

Let α, β, and γ be the roots of $x^3 + ax^2 + bx + c = 0$.

Then, we get

$$\sum_1 = \alpha + \beta + \gamma = -a, \quad \ldots.(1)$$

$$\sum_2 = \alpha\beta + \beta\gamma + \gamma\alpha = b, \quad \ldots.(2)$$

$$\sum_3 = \alpha\beta\gamma = -c. \quad \ldots.(3)$$

We have to form the equation whose roots are α^2, β^2, and γ^2.
Using (1), (2) and (3), we find the following:

$$\sum_1 = \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) = (-a)^2 - 2(b) = a^2 - 2b,$$

$$\sum_2 = \alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 = (\alpha\beta + \beta\gamma + \gamma\alpha)^2 - 2((\alpha\beta)(\beta\gamma) + (\beta\gamma)(\gamma\alpha) + (\gamma\alpha)(\alpha\beta))$$

$$= (\alpha\beta + \beta\gamma + \gamma\alpha)^2 - 2\alpha\beta\gamma(\beta + \gamma + \alpha) = (b)^2 - 2(-c)(-a) = b^2 - 2ca$$

$$\sum_3 = \alpha^2\beta^2\gamma^2 = (\alpha\beta\gamma)^2 = (-c)^2 = c^2.$$

Hence, the required equation is

$$x^3 - \left(\alpha^2 + \beta^2 + \gamma^2\right)x^2 + \left(\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2\right)x - \alpha^2\beta^2\gamma^2 = 0.$$

That is, $x^3 - \left(a^2 - 2b\right)x^2 + \left(b^2 - 2ca\right)x - c^2 = 0.$

Example 3.7

If p is real, discuss the nature of the roots of the equation $4x^2 + 4px + p + 2 = 0$, in terms of p.

Solution

The discriminant $\Delta = (4p)^2 - 4(4)(p+2) = 16\left(p^2 - p - 2\right) = 16(p+1)(p-2)$. So, we get

$$\Delta < 0 \text{ if } -1 < p < 2$$

$$\Delta = 0 \text{ if } p = -1 \text{ or } p = 2$$

$$\Delta > 0 \text{ if } -\infty < p < -1 \text{ or } 2 < p < \infty$$

Thus the given polynomial has

imaginary roots if $-1 < p < 2$;

equal real roots if $p = -1$ or $p = 2$;

distinct real roots if $-\infty < p < -1$ or $2 < p < \infty$.

EXERCISE 3.1

1. If the sides of a cubic box are increased by 1, 2, 3 units respectively to form a cuboid, then the volume is increased by 52 cubic units. Find the volume of the cuboid.
2. Construct a cubic equation with roots
   (i) $1, 2$, and 3 (ii) $1, 1$, and -2 (iii) $2, \frac{1}{2}$ and 1.
3. If α, β and γ are the roots of the cubic equation $x^3 + 2x^2 + 3x + 4 = 0$, form a cubic equation whose roots are
   (i) 2α, 2β, 2γ (ii) $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$ (iii) $-\alpha, -\beta, -\gamma$
4. Solve the equation $3x^3 - 16x^2 + 23x - 6 = 0$ if the product of two roots is 1.
5. Find the sum of squares of roots of the equation $2x^4 - 8x^3 + 6x^2 - 3 = 0$.

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